

AI Calorie Estimation Tutorial

Before starting, please complete the basic configuration and confirm that Jetson CSI camera, vision model, and OpenClaw dialogue channel can be used normally. This article only expands on the diet calorie estimation application itself.

Functions in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

1. Tutorial Objectives

This article demonstrates the vision application capability of OpenClaw on Jetson through "taking photos to recognize food + estimating calories + OLED/RGB/servo feedback". After completion, you can achieve:

- Take photos to observe current food.
- Estimate food calories, calories, and main nutritional information.
- OLED displays the recognized food name.
- Control RGB color based on calorie range.
- Control servo to indicate whether consumption is recommended based on whether it exceeds 1000 calories.

2. Hardware Preparation

- Jetson Nano B01
- OpenClaw expansion board
- CSI camera
- RGB light strip
- Optional: 1 or more servos
- Optional: AI voice interaction module + speaker
- Wiring diagram



```
~/run_usb_docker.sh
```

This project mainly uses:

```
/usr/local/bin/dht --format json # Get sensor data
/usr/local/bin/dht_oled --format json # Display sensor data on OLED
/usr/local/bin/rgb --format json color --color red # Turn all lights red
/usr/local/bin/rgb --format json effect --effect breathe --color blue --speed 5 # Blue
breathing light
/usr/local/bin/servo call-tool Servo1 --action set_angle --angle 90 # Servo1 turns to 90
degrees
/usr/local/bin/oled text --text 'Hello OpenClaw'
/usr/local/bin/oled clear
/usr/local/bin/csi_snap snap
/usr/local/bin/csi_snap snap --width 1280 --height 720
```

Calorie judgment rules suggestions:

- 0 to 500 calories: RGB lights up green.
- 500 to 1000 calories: RGB lights up yellow.
- Above 1000 calories: RGB lights up red.
- Exceeds 1000 calories: Servo turns to 90 degrees, indicating consumption not recommended.
- Does not exceed 1000 calories: Servo turns to 0 degrees, indicating consumption recommended.

4. Basic Capability Test

Enter TUI first:

```
openclaw tui

root@jetson-yahboom:~# openclaw tui
🔊 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
█
```

It is recommended to test in order:

- Take a photo
- OLED displays "Ready"
- RGB lights up blue

After individual capabilities are normal, test "food calorie detection".

5. Three Dialogue Methods

Method	Suitable Scenario	Suggested Positioning
TUI	Local debugging, observing model output	Preferentially used to verify vision recognition and calorie estimation
WhatsApp	Mobile remote diet recording	Suitable for demonstrating photo return results
Voice	Desktop life assistant	Suitable for triggering complete process with short sentences

5.1 WhatsApp Method

You can send:

- Food calorie detection
- Take a photo to check current food, estimate calories and nutritional information
- See how many calories this food has approximately
- Take a photo to see current food, estimate the food calories. If it exceeds 1000 calories, light up red light, otherwise light up green light. Servo rotates to 90 degrees to indicate not recommended for consumption, 0 degrees indicates recommended for consumption

20:23



< 2



+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current food, estimate the food calories. If it exceeds 1000 calories, light up red light; otherwise light up green light. servo rotates to 90 degrees to indicate not recommended for consumption, 0 degrees indicates recommended for consumption.

20:22 ✓✓



I can see a **Coca-Cola can** in the image. Let me estimate the calories and provide the appropriate feedback.

Calorie Estimate: A standard 330ml Coca-Cola can contains approximately **140-150 calories**.



Since this is well under 1000 calories



5.2 TUI Method

Run:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🚩 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

Recommended input:

- Take a photo to check current food, estimate food calories, calories, and nutritional information
- Food calorie detection, if it exceeds 1000 calories light up red and turn servo to 90 degrees



5.3 Voice Method

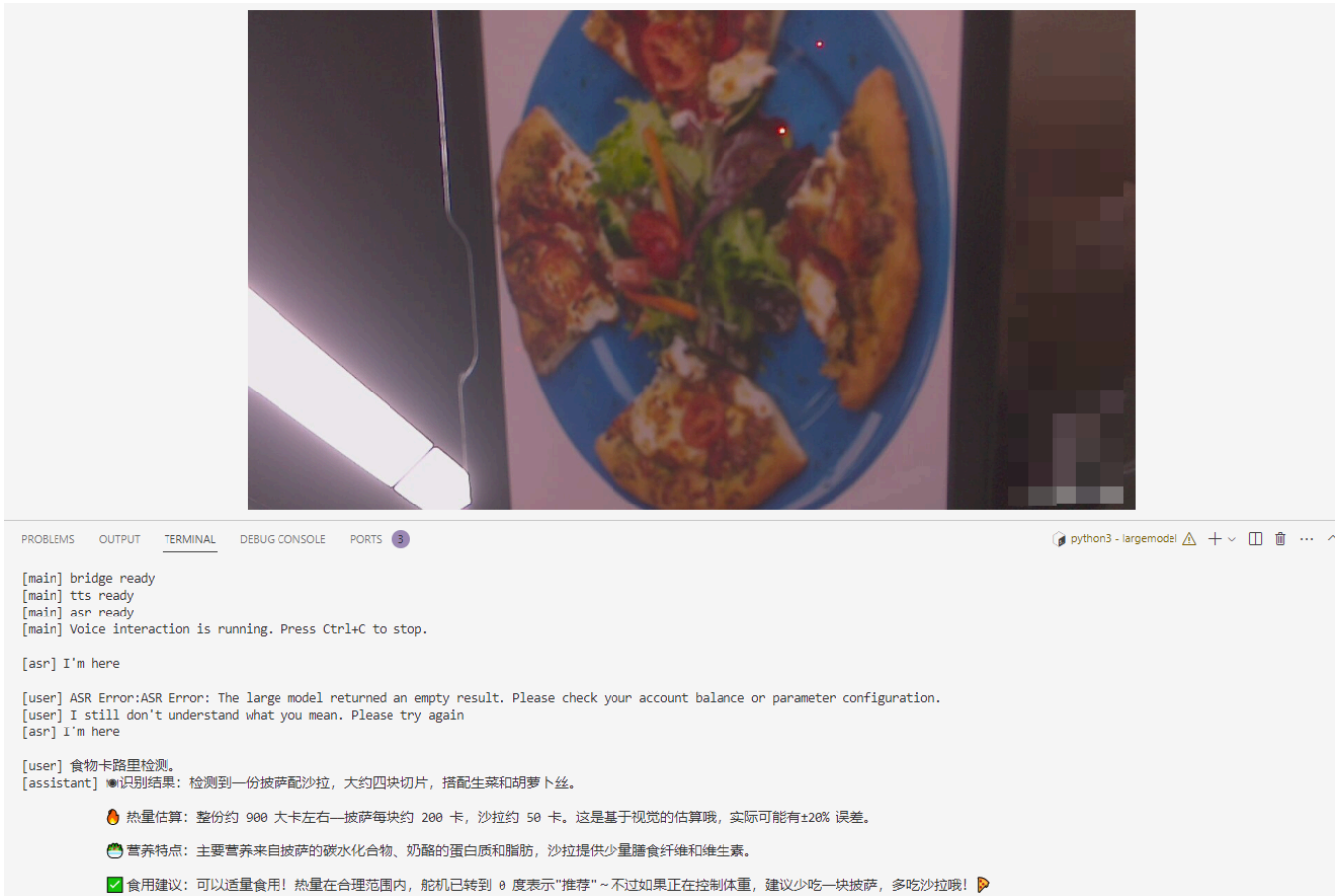
Execute the following command with openclaw gateway running:

```
cd /root/yahboom_ws/src/largemod1
python3 main.py
```

You can say:

- Food calorie detection
- Check how many calories this food has for me

- See if this food is suitable to eat now



6. Comprehensive Cases

6.1 Food Photography

Experiment Objective: Get current food photo to prepare for subsequent estimation.

Example Commands:

- Take a food photo for me
- Take a photo of the current plate
- Take a new food photo again

6.2 Food Recognition

Experiment Objective: Recognize the main food or drink in the image.

Example Commands:

- See what food this is
- Recognize what's mainly on the plate
- List the main ingredients

6.3 Calorie Estimation

Experiment Objective: Estimate total calories and explain the main nutritional sources.

Example Commands:

- Food calorie detection
- Estimate how many calories this food has
- Provide food name, calories, and main nutritional information

6.4 Peripheral Feedback

Experiment Objective: Feedback the estimation results through RGB, OLED, and servo.

Example Commands:

- Food calorie detection, OLED displays food name
- If below 500 calories light up green, 500 to 1000 calories light up yellow, above 1000 calories light up red
- If exceeds 1000 calories servo turns to 90 degrees, otherwise servo turns to 0 degrees

7. Common Problems

- Only takes photo without estimation: Check if the `camera` tool's describe capability or vision analysis process was called.
- Calorie answer too short: Need to explicitly include "estimate calories, calories, nutritional information, and give a suggestion" in the command.
- Peripheral has no action: Check if RGB and servo actions were generated in the reply, don't just look at the natural language conclusion.
- Estimation not accurate: This is vision estimation, suitable for demonstration and reference, not suitable as medical or nutritionist-level accurate results.